

SHENGYANG TAOXIAN, China

WMO: 543424

Lat: 41.640N

Lon: 123.483E

Elev: 60

StdP: 100.60

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.9	-21.2	-29.0	0.3	-20.7	-27.0	0.3	-18.6	9.3	-3.5	8.2	-4.2	1.7	140	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	32.1	23.3	30.9	23.1	29.9	22.5	26.1	29.7	25.3	28.8	24.4	27.7	4.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.3	28.8	24.1	19.1	27.7	23.1	18.0	26.6	81.6	29.9	77.5	28.6	74.1	27.3	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.0	8.5	7.3	-26.7	34.2	3.3	1.5	-29.1	35.3	-31.0	36.2	-32.9	37.1	-35.3	38.2		
			-26.9	27.4	3.3	0.9	-29.3	28.0	-31.2	28.6	-33.0	29.1	-35.4	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-11.6	-6.5	1.6	10.6	17.7	22.3	25.0	23.9	18.3	10.1	0.3	-8.6	(6)
(7)		DBStd	13.42	5.23	5.72	5.07	4.55	3.51	2.59	2.02	2.50	3.28	4.69	5.96	5.65	(7)
(8)		HDD10.0	2372	669	462	264	47	1	0	0	0	1	57	295	577	(8)
(9)		HDD18.3	4106	928	695	520	233	53	3	0	1	40	256	541	836	(9)
(10)		CDD10.0	1886	0	0	3	66	240	368	466	430	249	61	4	0	(10)
(11)		CDD18.3	578	0	0	0	2	35	121	207	173	39	1	0	0	(11)
(12)		CDH23.3	4717	0	0	0	34	368	962	1695	1328	317	13	0	0	(12)
(13)		CDH26.7	1418	0	0	0	3	88	307	562	407	50	1	0	0	(13)
(14)	Wind	WSAvg	3.3	2.7	3.2	3.8	4.4	4.1	3.3	2.8	2.6	2.8	3.3	3.5	2.8	(14)
(15)	Precipitation	PrecAvg	665	6	6	16	32	49	101	183	147	61	36	20	11	(15)
(16)		PrecMax	917	27	25	36	69	111	164	328	362	140	107	48	22	(16)
(17)		PrecMin	490	0	0	0	4	16	40	85	39	2	4	4	2	(17)
(18)		PrecStd	124	6	8	10	19	27	34	73	82	31	25	13	6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	10.1	18.8	26.9	31.2	33.2	34.0	33.1	29.9	25.0	18.0	7.8	(19)
(20)			MCWB	0.4	3.6	10.0	14.5	17.5	20.5	25.0	25.0	21.8	16.5	11.5	4.6	(20)
(21)		2%	DB	1.9	7.0	14.9	23.9	29.0	31.2	32.1	31.2	28.1	22.8	14.2	4.8	(21)
(22)			MCWB	-1.3	2.3	7.1	12.9	16.9	20.0	24.8	24.6	19.7	15.0	9.1	1.7	(22)
(23)		5%	DB	-0.1	4.9	12.1	21.2	27.1	30.0	31.0	30.1	26.9	20.2	12.1	2.8	(23)
(24)			MCWB	-2.7	0.9	5.6	11.4	15.9	19.8	24.4	23.9	18.7	13.5	7.7	0.2	(24)
(25)		10%	DB	-2.2	2.9	9.9	19.1	25.1	28.2	29.8	29.0	25.1	18.1	9.8	0.8	(25)
(26)			MCWB	-4.4	-0.4	4.1	10.1	14.8	19.7	23.9	23.2	17.8	12.2	6.3	-1.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	4.8	10.5	16.2	20.2	23.6	27.5	27.2	23.5	17.8	12.2	4.8	(27)
(28)			MCDWB	3.1	8.1	17.4	23.4	28.6	29.6	31.7	30.7	27.8	22.9	16.8	7.3	(28)
(29)		2%	WB	-1.1	2.7	8.0	14.0	18.5	22.5	26.4	26.1	21.7	15.9	10.1	2.0	(29)
(30)			MCDWB	1.2	6.1	13.1	21.6	25.7	28.0	30.4	29.4	25.5	20.7	13.6	3.9	(30)
(31)		5%	WB	-2.8	1.1	6.2	12.5	17.3	21.6	25.5	25.4	20.5	14.7	8.3	0.1	(31)
(32)			MCDWB	-0.4	4.2	11.5	19.9	24.4	26.8	29.3	28.7	24.5	19.1	11.6	2.1	(32)
(33)		10%	WB	-4.6	-0.5	4.5	11.0	16.2	20.9	24.7	24.5	19.3	13.2	6.1	-1.6	(33)
(34)			MCDWB	-2.5	2.5	9.1	17.6	23.1	26.1	28.2	27.7	23.3	17.2	9.2	0.4	(34)
(35)	Mean Daily Temperature Range	MDBR	12.1	12.0	10.9	12.2	11.9	10.0	8.4	9.0	12.0	12.0	10.2	10.9	(35)	
(36)		5% DB	MCDBR	13.1	12.7	13.9	15.3	14.5	12.9	10.4	10.2	13.3	13.6	12.4	11.4	(36)
(37)			MCWBR	10.8	9.0	8.4	7.8	6.7	5.4	4.8	4.9	6.3	7.5	8.6	9.4	(37)
(38)		5% WB	MCDWB	12.7	11.7	12.8	13.4	12.1	10.0	8.6	8.2	10.9	11.8	11.2	10.7	(38)
(39)			MCWBR	10.8	8.8	8.2	7.5	6.2	5.0	4.8	4.6	6.0	7.3	8.6	9.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.374	0.441	0.499	0.600	0.650	0.727	0.706	0.618	0.541	0.551	0.465	0.396	(40)	
(41)		taud	2.014	1.864	1.802	1.650	1.605	1.529	1.606	1.766	1.867	1.761	1.892	1.994	(41)	
(42)		Ebn at Noon	722	724	734	691	671	621	629	672	691	606	607	652	(42)	
(43)		Edh at Noon	118	161	193	241	259	280	256	212	179	178	134	112	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.36	3.21	4.30	5.09	5.78	5.28	4.82	4.72	4.40	3.28	2.26	1.91	(44)	
(45)		RadStd	0.15	0.21	0.28	0.33	0.44	0.53	0.39	0.38	0.26	0.21	0.22	0.14	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) -346	(k) -344	(l) N/A	(m) N/A	(46)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page